Data Is the First Step

Use the project datasets to understand measurement types, data structures, file formats, ingestion patterns, and reproducible notebook/repository workflows.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 17–48. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.



Detailed Notes

The book's project datasets

Source: book pp. 17–23 / PDF p. 33

The examples span retail price, healthcare outreach, energy use/production, advertising sales, financial fraud, telecommunications churn, and used-car auctions. Each dataset teaches a different target and workflow; names and sample values do not remove the need to inspect actual columns, units, labels, and provenance.

Use a data dictionary and business context. Ask what one row represents, how the label is produced, when features become available, and whether records are independent. Confirm licensing and sensitive information before uploading to a cloud notebook or repository.

Quantitative, qualitative, and data structures

Source: book pp. 23–25 / PDF p. 39

Quantitative values can be continuous or discrete; qualitative values describe categories, labels, or text. Encodings can obscure meaning: an integer category code is not necessarily a numeric magnitude.

Structured data follows a schema, often tabular. **Semi-structured** data has nested/key-value organization such as JSON. **Unstructured** data includes free text, images, and audio. Each needs different parsing, validation, and feature construction. Missing, invalid, and unknown values need distinct handling where they mean different things.

File types and processing modes

Source: book pp. 25–27 / PDF p. 41

CSV is portable but weakly typed; inspect delimiter, quoting, encoding, null markers, and date parsing. JSON supports nesting; Parquet and similar columnar formats support efficient typed analytics; image/audio files need metadata and label alignment.

Batch data arrives in collections or scheduled snapshots; **streaming** data arrives continuously. Their latency, ordering, state, and validation needs differ. Avoid reading only the first few rows and assuming the full file is clean. Validate row counts, schema, duplicates, and date coverage.

GitHub data repository and Colab

Source: book pp. 27–48 / PDF p. 43

A repository can version notebooks, documentation, small sample data, and transformation code. Large/private production data may require separate storage, access controls, or data-versioning tools; never commit secrets or restricted records.

Colab provides a hosted notebook environment, but runtime state is temporary. Mounting cloud drives and downloading datasets require explicit permissions. Keep notebooks restartable: record packages, inputs, seeds, and steps; use relative/configurable paths; clear sensitive outputs; and rerun from a clean runtime before sharing.

Revision Check

1. Which columns are measurements and which are categories?
2. Is this data batch or streaming?
3. What should not be committed to GitHub?
4. Can the notebook reproduce its results from a clean runtime?

Data format, schema, provenance, and ingestion are part of the model. Inspect before transformation and make the analysis reproducible.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.